The Beneficial Effect of Angiotensin-Converting Enzyme Inhibition With Captopril on Diabetic Nephropathy in Normotensive IDDM Patients With Microalbuminuria

Lori M.B. Laffel, MD, MPH, *Boston, Massachusetts*, Janet B. McGill, MD, *St. Louis, Missouri*, Daniel J. Gans, MS, *Princeton, New Jersey*, on behalf of the North American Microalbuminuria Study Group*

PURPOSE: To determine whether angiotensin-converting enzyme (ACE) inhibition with captopril reduces the progression of microalbuminuria to overt proteinuria in normotensive patients with insulin-dependent diabetes mellitus (IDDM).

PATIENTS AND METHODS: This study was a prospective randomized, double-blind, placebo-controlled trial involving 26 centers in the United States and Canada. One hundred forty-three subjects, 14 to 57 years of age, with IDDM for 4 to 33 years, blood pressure <140/90 mm Hg in the absence of antihypertensive therapy, and persistent albumin excretion 20 to 200 μg/min were randomized to double-blind treatment with captopril 50 mg or placebo BID. Albumin excretion rate (AER), blood pressure, and glycohemoglobin were determined every 3 months, and creatinine clearance (CrCl) and urea excretion were measured every 6 months.

RESULTS: Within 24 months, 6.0% (4/67) of captopril-treated subjects and 18.6% (13/70) of placebo-treated subjects progressed to clinical proteinuria, defined as AER >200 μg/min and at least 30% above baseline (risk reduction = 67.8%, *P* = 0.037). AER increased at an annual rate of 11.8% (95% confidence interval [CI] −3.3% to 29.1%) in the placebo group, while it declined by 17.9% (CI −29.6% to −4.3%) in the captopril group (*P* = 0.004). CrCl decreased by 4.9 mL/min per 1.73 m² per year in the placebo group, while it remained stable in the captopril group (0.9 mL/min per 1.73 m² per year, *P* = 0.039 between groups). Ten subjects required treatment for hypertension; 8 in the placebo group and 2 in the captopril group. There was little correlation between the 24-month changes in mean arterial blood pressure and AER in either group. Glycohemoglobin and urinary urea excretion did not differ between groups.

CONCLUSIONS: After 24 months of therapy with captopril, compared with placebo, normotensive subjects with IDDM experienced significantly less progression of microalbuminuria to clinical proteinuria, reduced albumin excretion, and preserved CrCl rate. The ACE inhibitor, captopril, was well tolerated.

From the Epidemiology and Genetics Unit (LMBL), Research Division, Youth Unit, Clinic Division, Joslin Diabetes Center, Harvard Medical School, Boston, Massachusetts; Division of Endocrinology (JBM), Department of Medicine, Washington University, St. Louis, Missouri; Bristol-Myers Squibb Co. (DJG), Princeton, New Jersey.

*Participating centers and investigators in the North American Microalbuminuria Study Group are listed in the Appendix.

This study was supported by a grant from the Bristol-Myers Squibb Institute for Pharmaceutical Research, Princeton, New Jersey. They did not have access to unblinded data.

Requests for reprints should be addressed to Dr. Laffel, Joslin Diabetes Center, One Joslin Place, Boston, Massachusetts 02215.

Manuscript submitted November 10, 1994 and accepted in revised form February 23, 1995.

Recent advances in the detection and treatment of diabetic nephropathy as well as improvements in general diabetes care have bettered the outlook for patients with insulin-dependent diabetes mellitus (IDDM).[1] Diabetic nephropathy can be found early in its course through the detection of microalbuminuria.[2-6] Interventions aimed at improved glycemic and blood-pressure control[7-11] and specific treatment with angiotensin-converting enzyme (ACE) inhibitors[12-16] may postpone and possibly prevent further increases in urinary albumin excretion and deterioration in kidney function.

While these interventions aimed at the secondary prevention of diabetic nephropathy are encouraging, the results of the Diabetes Control and Complications Trial (DCCT)[8] and other studies[9,17] suggest that primary prevention of diabetic nephropathy may be possible in some patients through intensive insulin therapy. It is unrealistic, however, to expect all patients with IDDM to attain near-normal glycemia. Indeed, the DCCT has shown that near normoglycemia remained an elusive goal for the majority of patients, with less than 5% of the study population attaining average glycohemoglobin (HbA_{1c}) levels within the normal range.[8]

Accordingly, therapies other than those directed at improved glycemic control are necessary to prevent or postpone diabetic nephropathy in the overwhelming majority of patients with IDDM to reduce the tremendous burden of this complication, which is the leading cause of end-stage renal disease in the western world. To date, interventions introduced during the stage of overt proteinuria[10,11,15] and during the earlier stage of microalbuminuria[13,14,16] have successfully reduced progressive kidney disease and renal failure.

It has recently been shown that, in addition to its hypotensive effect, the ACE inhibitor, captopril, pos-

sesses specific renal protective effects, as demonstrated by its ability to reduce by 50% the progression to end-stage renal disease (dialysis or transplantation) or death in a study of over 400 proteinuric patients with IDDM followed up for an average of 3 years.[15] We now report the effect of the same ACE inhibitor, captopril, on progression from microalbuminuria to proteinuria in the largest randomized, double-blind, placebo-controlled trial of normotensive patients with IDDM and microalbuminuria followed up for 2 years.

PATIENTS AND METHODS

Thirty investigators from 26 centers in North America participated in the 24-month prospective, randomized, double-blind, placebo-controlled trial in accordance with standardized procedures outlined in the study protocol. Investigators complied with their own institutional review boards, and all subjects gave written informed consent.

The study consisted of a screening visit, a 3-week, single-blind, placebo lead-in period, and a 24-month, double-blind, placebo-controlled treatment period. Patients between the ages of 14 and 57 years with 4 to 33 years of documented IDDM diagnosed before 45 years of age were eligible for study participation if their overnight urinary albumin excretion rate was 20 to 200 µg/min. IDDM was confirmed by a history of ketonuria and the need for continuous insulin therapy, except for periods lasting less than 6 months during the first 2 years after diagnosis.

Criteria for exclusion were as follows: HbA_{1c} levels ≥11.5%; body weight outside of 75% to 125% of ideal; serum creatinine and potassium levels beyond the normal ranges; white blood cell count <3,500/mm^3; blood pressure (systolic/diastolic) ≥140/90 mm Hg; antihypertensive therapy; pregnancy; lactation; or failure to use adequate contraception for women of child-bearing ages. Histories of renal, cardiac, hepatic, gastrointestinal, or autoimmune diseases were also causes for exclusion. Use of calcium channel blockers, ß-blockers, and nonsteroidal anti-inflammatory agents (except for low-dose aspirin not to exceed 650 mg/d) was prohibited.

During the placebo lead-in period, three additional overnight urine samples were requested for albumin excretion rate (AER) measurements. If 2 out of 3 AERs (occasionally including the screening urine sample) were between 20 and 200 µg/min, subjects had a full history, physical, electrocardiogram, and laboratory evaluation for confirmation of eligibility status. Heart rate and blood pressure (Korotkoff Method I/V by random-zero mercury sphygmomanometer with an appropriately sized cuff) were obtained according to standardized methods after the subject had been seated for 5 minutes. Blood pressure was measured twice at 30-second intervals, followed by a standing blood pressure measurement after 2 minutes. The average of the two sitting measurements was used, and mean arterial blood pressure was diastolic pressure plus one third of the pulse pressure.

After eligibility was confirmed, subjects were randomized in blocks of two according to center to receive captopril 50 mg or placebo BID. During the 24-month treatment period, subjects were reexamined every 3 months in addition to a single visit 2 weeks after randomization. Two overnight urinary AER samples, blood-pressure measurements, and laboratory profiles were obtained at each of these visits (except for the 2-week visit, which included only blood pressure measurements and laboratory profiles). In addition, 24-hour urine samples were collected at baseline and every 6 months for creatinine clearance (CrCl) and urinary urea nitrogen determinations. Overnight and 24-hour urine samples were recollected if creatinine excretion indicated an incomplete collection. Compliance, monitored by the pill-count method, was defined as ingestion of between 80% and 110% of the pills provided.

During the active-treatment period, diet and insulin management of diabetes were not altered, unless diet exceeded the recommended 50% to 60% carbohydrate, 30% fat, and 20% protein guideline, or HbA_{1c} levels exceeded 11.5%. Diet evaluation was performed at months 0, 6, 12, 18, and 24 of the trial. If blood pressure was ≥140/90 mm Hg at two consecutive visits during active treatment, the subject received prazosin or clonidine. Therapy was titrated according to standard medical practice.

All blood and urine samples were analyzed at a central laboratory (SciCor, Inc., Indianapolis, Indiana) according to standard methods. Blood and urine chemistries were performed on a multichannel autoanalyzer. HbA_{1c} measurements were performed by two different methods confirmed to give comparable results (originally performed manually by column [Bio-RAD] and subsequently by automated methods using ion-exchange high-performance liquid chromatography [DIAMAT]). Urinary albumin concentrations were measured in duplicate from fresh urine samples by radioimmunoassay (double-antibody iodine 125 [^{125}I] radioimmunoassay by Diagnostics Products, Corp., Los Angeles, California).

Measures of Efficacy

The primary outcome variable was progression to clinical proteinuria, defined as AER >200 µg/min and at least 30% above the baseline AER on two consecutive visits. The change from baseline AER and the annual rate of change of AER were also assessed. Other outcomes included the change from baseline

and the rate of change of CrCl and serum creatinine and progression to hypertension. Confounding variables such as HbA_{1c} levels (glycemic control), urea excretion (protein intake), and blood pressure were examined for their effects on AER.

Measures of Safety

Treatment-emergent effects were solicited and recorded at each study visit by the research nurse. Attribution was determined by the investigator at each site.

Statistical Analysis

Results were analyzed with Statistical Analysis System (SAS) software (SAS/STAT Software, Release 6.07. SAS Institute, Inc., Cary, North Carolina). Applying the intention-to-treat principle, all randomized patients, so long as they had the necessary data, were included in all analyses.

The last three pretreatment AER measurements were averaged to give the mean baseline AER, which was then log transformed. For each quarterly follow-up visit during double-blind treatment, two overnight urine samples were collected for AER determination. The mean of the two was computed and again log transformed. Logs were used for analysis because the AER values were positively skewed. AER results are presented in terms of the corresponding geometric means.

Demographics and baseline variables were compared between treatment groups by Fisher's exact test for categoric variables and by single-factor analysis of variance for continuous variables. Analysis of progression was performed using Fisher's exact test to compare the proportion of progressors between treatment groups. Time to progression was analyzed by life-table methods. The Kaplan-Meier method[18] was used to estimate cumulative progression rate curves, which were compared statistically with the Cox-Mantel log-rank test.[19] Risk reduction was determined with the Cox proportional hazards regression model.

Cross-sectional analyses were performed for each visit time on AER, CrCl, blood pressure, and HbA_{1c} data by the method of analysis of covariance. The variable analyzed was change from baseline, and the baseline itself served as the covariate.

For rate-of-change analyses, all patients with a baseline and at least one double-blind reading were used. For AER, these readings were log-transformed means, as noted previously. Individual regression slopes were determined from the variable-versus-time data. Mean slopes for the treatment groups were obtained and compared using a mixed model (SAS procedure MIXED), which weights individual slopes in accordance with their respective precisions.

TABLE I

Baseline Characteristics of Subjects Randomized to Captopril or Placebo

Baseline Characteristic	Captopril (n = 70)	Placebo (n = 73)
Sex (%)		
Male	53	48
Female	47	52
Age (y)	32.0 ± 8.1	33.4 ± 9.0
Duration of diabetes (y)	18.3 ± 6.6	18.3 ± 5.9
Body weight (kg)	70.9 ± 12.0	72.1 ± 11.6
AER (μg/min)	62 ± 36	62 ± 41
CrCl (mL/min per 1.73 m^2)	80 ± 21	80 ± 22
Serum creatinine (mg/dL)	1.1 ± 0.2	1.1 ± 0.2
HbA_{1c} (%)	7.6 ± 1.5	8.0 ± 1.3
MAP (mm Hg)	92 ± 6	92 ± 7
Urinary urea nitrogen (g/24 h)	10.7 ± 3.8	10.7 ± 4.3

Data are presented as mean ± standard deviation; geometric mean of AER is shown. No differences between groups are significant (P >0.10 in all cases).
AER = albumin excretion rate; CrCl = creatinine clearance; HbA_{1c} = glycohemoglobin; MAP = mean arterial pressure.

Results are expressed as mean change (in the case of AER, geometric mean percentage change) per year.

Relationships between changes from baseline in any two variables were assessed with Spearman's rank-order correlation method. Results are presented as mean ± standard deviation unless otherwise stated. All P values are two sided. A P value ≤0.05 was considered to indicate statistical significance.

RESULTS

Study Population

Two hundred fifteen subjects were enrolled during the screening period, from which 143 fulfilled eligibility criteria and were randomized, 70 to captopril treatment and 73 to placebo. Of the 72 subjects screened but not randomized, 60 did not qualify and 12 elected not to participate.

The 143 subjects had a mean age of 32.7 years (range 14 to 57); 72 (50.3%) were male, and 131 (91.6%) were white. Their mean duration of diabetes was 18.3 years (range 4 to 33), and their mean age at onset was 13.8 years (range 1 to 44). One hundred two subjects (71.3%) were nonsmokers. Subjects received an average of 50.4 U/d of insulin.

Table I shows the comparability of demographic and baseline characteristics of study subjects randomized to captopril and placebo. Sex distribution, age, race, proportion of smokers, and duration of diabetes were similar between groups. In addition, the baseline AER, CrCl, serum creatinine, HbA_{1c}, mean arterial pressure, and 24-hour urinary urea nitrogen excretion were similar. The geometric means of the AERs were identical at 62 μg/min (range 18 to 283) in the captopril and placebo groups. All subjects had persistent microalbuminuria. Overall, compliance

TABLE II

Efficacy of 24 Months of Captopril Versus Placebo Therapy* in Normotensive Patients With IDDM

Outcome	Captopril	Placebo
Progression to clinical proteinuria	4/67 (6.0%)	13/70 (18.6%)†
AER		
% change/y (slope)	-17.9%	11.8%‡
24-month adjusted mean change from baseline§	-42.4%	13.5%†
CrCl (mL/min per 1.73 m²)		
Change/y (slope)	0.9	-4.9†
24-month adjusted mean change from baseline§	4	-8†
Progression to hypertension	2/70 (3%)	8/73 (11%)

*Captopril 50 mg or placebo BID.
†$P \leq 0.05$ between groups.
‡$P \leq 0.01$ between groups.
§Adjusted by analysis of covariance.
IDDM = insulin-dependent diabetes mellitus; AER = albumin excretion rate; CrCl = creatinine clearance.

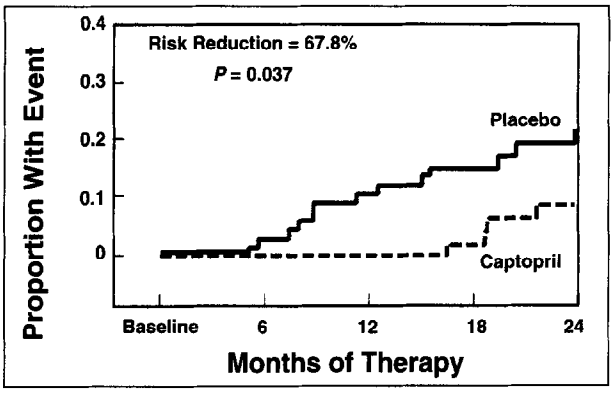

Figure 1. Probability of progression to clinical proteinuria in 67 captopril-treated subjects (**dashed line**) and 70 placebo-treated subjects (**solid line**) during 24 months of treatment (captopril 50 mg or placebo BID) in normotensive patients with insulin-dependent diabetes mellitus.

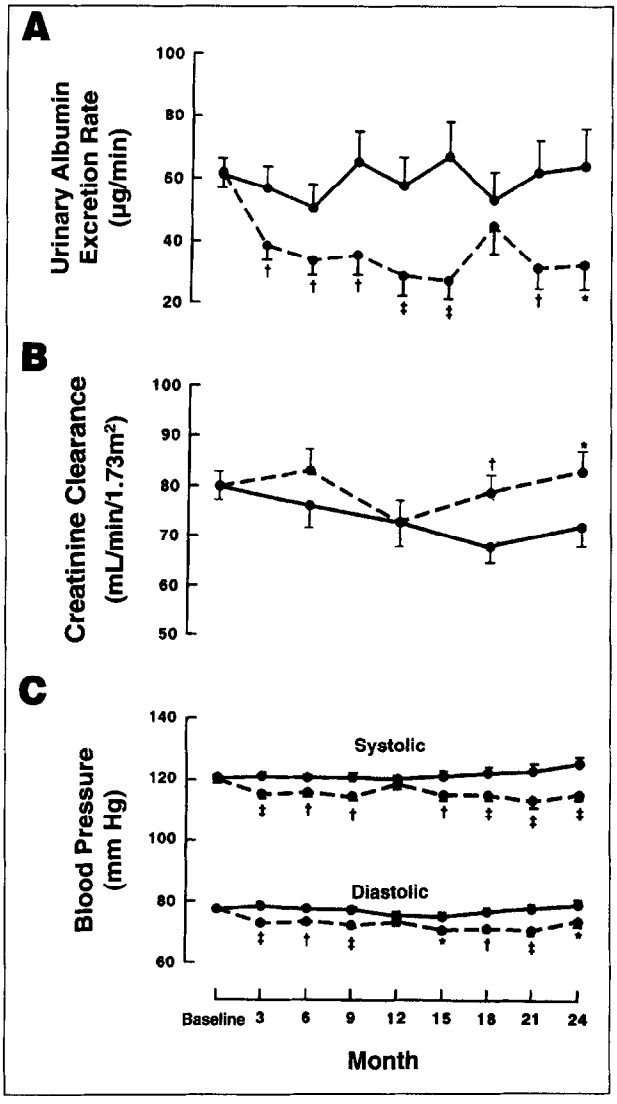

Figure 2. Effect of 24 months of captopril (50 mg BID; **dashed line**) compared to placebo treatment (BID; **solid line**) on **A.** urinary albumin excretion rate, **B.** creatinine clearance, and **C.** blood pressure in normotensive patients with insulin-dependent diabetes mellitus. Significant difference between groups: *$P \leq 0.05$; †$P \leq 0.01$; ‡$P \leq 0.001$. Data reported as mean ± standard error of the mean for urinary albumin excretion rate and creatinine clearance, and mean for blood pressure.

was similar between groups (79% captopril, 68% placebo).

Forty-three subjects discontinued double-blind treatment during the 24-month study period for the following reasons: 19 were lost to follow-up (11 captopril, 8 placebo); 7 had an adverse drug experience (4 captopril, 3 placebo); 5 asked to discontinue study participation (3 captopril, 2 placebo); 3 were noncompliant (2 captopril, 1 placebo); 2 had a concomitant event (placebo); 2 violated protocol (1 captopril, 1 placebo); 2 had uncontrolled hypertension (placebo); 1 became pregnant (placebo); 1 required therapy prohibited in the protocol (placebo); and 1 died accidentally (motor vehicle accident) (captopril).

Efficacy

Follow-up AER data were available on 67 captopril and 70 placebo patients. The numbers of patients progressing from microalbuminuria to clinical proteinuria are shown in **Table II**. For placebo, 18.6% (13/70) progressed, while 6% (4/67) progressed to proteinuria during captopril therapy. The proportions progressing were significantly different (Fisher's exact test $P = 0.037$) as was the time to progression (log-rank test $P = 0.037$, **Figure 1**). Over 2 years, captopril treatment was associated with a 67.8% risk reduction in progression from microalbuminuria to overt proteinuria. Progressors first appeared after 6 months of placebo treatment, while the first progressor appeared after 17 months in the captopril group.

At all visits, AER was lower than baseline in the captopril group, while it was higher than baseline at most visits in the placebo group (**Figure 2A**). After 24 months, AER had decreased by 42.4% (± 11% stan-

dard error of the mean [SEM]) in the captopril group, while it had increased by 13.5% (± 19.8%) (SEM) in the placebo group ($P = 0.004$). AER increased at an annual rate of 11.8% (95% confidence interval [CI] –3.3% to 29.1%) in the placebo group, while the annual rate of decline was 17.9% (CI –29.6% to –4.3%) in the captopril group. The difference between mean slopes was significant ($P = 0.004$).

There were significant differences in 24-hour CrCl levels between the captopril and placebo groups (**Figure 2B**). CrCl decreased in the placebo group from 81 ± 3 to 72 ± 4 mL/min per 1.73 m^2 (mean ± SEM) after 24 months of double-blind treatment. In contrast, CrCl increased slightly from baseline in the captopril group from 79 ± 3 to 83 ± 4 mL/min per 1.73 m^2. This difference was significant between the groups after 24 months ($P = 0.033$). The mean rate of decline in CrCl in the placebo group was 4.9 mL/min per 1.73 m^2 per year (95% CI –8.7 to –1.1) ($P \leq 0.05$). In contrast, there was no significant change in CrCl over time in the captopril-treated group (0.9 mL/min per 1.73 m^2 per year (95% CI –3.1 to 4.9). Again, the difference between slopes was significant ($P = 0.039$). While captopril's effects on AER and CrCl appeared within the first 3 to 6 months, between months 6 and 24, differences between captopril and placebo remained. From month 6 to 24, CrCl in the placebo group changed, –4.2 mL/min per 1.73 m^2 per year versus 1.5 mL/min per 1.73 m^2 per year in the captopril group. Despite the change in CrCl, however, serum creatinine did not change in either the placebo or captopril group after 24 months of treatment.

Ten subjects progressed to hypertension and required treatment, 2 in the captopril group and 8 in the placebo group. According to protocol, hypertensive subjects received prazosin or clonidine. In addition, 4 of the patients received treatment with diuretics and 1 received doxazosin. Of those requiring antihypertensive therapy, 3 patients progressed to clinical proteinuria, all from the placebo group.

Captopril reduced systolic, diastolic, and mean arterial blood pressures. The decreases from baseline were statistically significant compared to placebo at most time points (**Figure 2C**). In the placebo group, a small, significant increase of 2.8 mm Hg in mean arterial blood pressure was observed at 24 months compared with baseline ($P \leq 0.05$). There was no significant correlation between the 24-month changes in mean arterial blood pressure and AER in either the captopril or placebo group ($r = .11$ for captopril, $r = .11$ for placebo). After adjustment for current blood pressure, a 53% risk reduction for progression from microalbuminuria to clinical proteinuria remained for the captopril group when compared with the placebo group (not significant [NS], $P = 0.21$).

Glycemic control did not differ between treatment groups at baseline or at any time points during the study. There was no correlation between the 24-month changes in HbA_{1c} and AER in either group. Urinary urea nitrogen excretion, a marker of protein intake, decreased from baseline to the 24-month point in both the captopril (–0.2 g/d) and placebo (–0.9 g/d) groups, but the difference between groups was not significant ($P = 0.43$). There was no correlation between the change in urinary urea nitrogen excretion at 24 months and the change in AER at 24 months in either group. Cholesterol levels were not significantly different between groups at any time during the study.

Safety

Of the 143 subjects randomized, 48 of 70 (68.6%) in the captopril group and 52 of 73 (71.2%) in the placebo group completed the 24-month follow-up. There were 9 subjects withdrawn from study treatment due to an adverse event, 4 of 70 (5.7%) in the captopril group (due to neutropenia, hemolytic anemia, orthostatic hypotension in a subject receiving spironolactone for hirsutism, and seizure aura in a subject with a known seizure disorder) and 5 of 73 (6.8%) in the placebo group (due to abnormal liver function tests in 2 subjects; vision disturbance, hypotension, and hypertensive crisis each in 1). The episode of neutropenia, which occurred in a 42-year-old man, was attributed directly to captopril treatment and resulted in a foot cellulitis that required IV antibiotics. The neutropenia resolved 1 month after discontinuing captopril. There was an accidental death by a motor vehicle accident in the captopril group unrelated to study treatment.

The groups experienced similar rates of treatment-emergent clinical events, 69 of 73 (94.5%) in the placebo group and 64 of 70 (91.4%) in the captopril group. There was no difference between adverse events except for headaches, which occurred more frequently in the placebo group (52.1% versus 34.3%, placebo versus captopril). Cough occurred equally in the 2 treatment groups, 15 of 73 (20.5%) in the placebo group and 16 of 70 (22.9%) in the captopril group, of which only 2 episodes were attributed to captopril treatment. A persistent increase from baseline in mean serum potassium level of 0.1 mEq/L was observed in the captopril group, which was not clinically significant. There were no episodes of hyperkalemia throughout the study.

COMMENTS

After 24 months of therapy with the ACE inhibitor, captopril, normotensive patients with IDDM had significantly less progression from microalbuminuria to overt proteinuria than patients treated with placebo ($P = 0.037$). The observed 24-month rate of progression of 6% (4 of 67) in the captopril-treated group com-

pared favorably with the 18.6% (13 of 70) rate of progression in the placebo group, a 67.8% risk reduction after 24 months of therapy ($P = 0.037$, by life-table analysis). Thus, over a 2-year period, for every 8 patients treated, progression to proteinuria will be prevented in 1.

During the 2 years of double-blind study, the mean percentage change from baseline AER differed significantly between groups at most times. While it rose in the placebo-treated group, it fell significantly in the captopril-treated group. In addition, the annual rate of change revealed an average increase of 11.8% per year with placebo compared to an average decrease of 17.9% per year with captopril, ($P = 0.004$ between groups).

Similarly, other outcomes were more favorable in the captopril group. Fewer patients needed treatment for hypertension, only 2 with captopril versus 8 with placebo. Systemic blood pressure decreased modestly from baseline by 3.5 mm Hg with captopril therapy, while it rose with placebo by 2.8 mm Hg at 24 months ($P \leq 0.05$). Since the level of blood pressure is strongly related to albumin excretion, it is possible that the observed changes resulted in part from the blood pressure differences between treatment groups.[20] The 24-month changes in blood pressure, however, were not significantly correlated with the changes in AER in either treatment group. After correcting for the changes in blood pressure, captopril treatment was associated with a 53% reduction in risk of progression from microalbuminuria to overt proteinuria (P = NS). Although the reduction in risk after blood pressure correction was not statistically significant, one possible explanation may be the small numbers of progressors (total of 17 in both groups).

Furthermore, the changes in AER could not be explained by other variables recognized to impact on albumin excretion, including glycemic control and dietary protein intake. There were no differences in either HbA_{1c} or urinary urea nitrogen excretion between treatment groups, and no correlation between these variables and changes in urinary albumin excretion.

In the placebo-treated group, the increases in AER were accompanied by a mean increase in blood pressure and a mean decrease in CrCl of 8 mL/min per 1.73 m^2. With captopril treatment for 24 months, AER was reduced and CrCl was preserved, which compared to previously published, smaller trials of ACE inhibition.[14,16,21,22] In addition to captopril's effect on systemic blood-pressure lowering, ACE inhibition appeared to have a specific renoprotective effect in the stage of incipient diabetic nephropathy, similar to its significant efficacy in preserving kidney function in patients with overt nephropathy, as shown in the Diabetic Nephropathy Collaborative Study.[15]

While there was no corresponding change in serum creatinine level, captopril treatment was associated with significant preservation of kidney function, as determined by CrCl, compared with placebo treatment. This is the first trial to document preservation of renal function with ACE inhibition initiated during the stage of microalbuminuria in IDDM patients. The mechanism of action of ACE inhibition in both the earlier stage of microalbuminuria and in the stage of overt nephropathy may be related to decreases in intraglomerular pressure[23-25] and subsequent preservation of glomeruli. While the decrement in AER after captopril therapy may result from changes in glomerular hemodynamics, Gansevoort et al[26] reported a dissociation of the acute renal hemodynamic effects and the antiproteinuric effects of ACE inhibition with enalapril in nondiabetic renal disease. They found the blood pressure and hemodynamic effects to be maximal within hours, while the maximal antiproteinuric effects were observed after 4 weeks of therapy.

Our trial included the largest number of normotensive IDDM patients with microalbuminuria, who had varying levels of renal function, based on 24-hour CrCl rates. It is unclear why this study population had lower CrCl at entry than other studies of patients with microalbuminuria.[16] We were unable to detect any assay bias, all urinary determinations were reviewed at the central laboratory prior to unblinding, all values were validated, and any found to be unreliable were excluded from analyses. It is possible that, due to the long duration of diabetes of these subjects, they may have surpassed the period of hyperfiltration and possessed various degrees of renal structural damage, as shown by Chavers et al.[27]

It is possible that ACE inhibition during the stage of microalbuminuria, and possibly even earlier, may provide a form of primary prevention in patients whose glomerular lesions may be more functional (ie, related to abnormal renal hemodynamics) than structural. ACE inhibition may be particularly helpful in correcting renal hemodynamic alterations induced by hyperglycemia, as reported by Jenkins et al.[28] They described normalization of renal hemodynamics (ie, reduced filtration fraction) after ACE inhibition with captopril in patients with the poorest glycemic control.[28] Since it may not be possible to improve glycemic control in the vast majority of patients with IDDM, future studies should examine alternative preventive interventions, such as blood-pressure reduction and specific renoprotection with ACE inhibition in patients with uncontrolled diabetes.[8,29,30] Compliance with oral therapy would be expected to be greater than that with any intensive diabetes treatment regimen. Furthermore, in cost-benefit analyses, ACE inhibition resulted in substantial cost savings when implemented early.[31,32] Finally, efforts to im-

prove glycemic control once persistent microalbuminuria exists do not appear to prevent progression to more advanced stages of albumin excretion, as recently shown.[33,34]

From our 24-month clinical trial, we cannot conclude that ACE inhibition with captopril prevents progression indefinitely. It may merely postpone progression by about 12 months (Figure 1) in patients who may possess fixed morphologic lesions. Follow-up studies are needed to clarify this point and to determine the rate of progression for all patients with microalbuminuria. The recent literature suggests that progression to proteinuria may occur in about 50% of IDDM patients with microalbuminuria,[35,36] a rate lower than the previously reported rate of over 80% found in earlier studies.[2-5]

Captopril's potential to prevent deterioration in renal function in patients with incipient nephropathy suggests that it may afford specific renal protection prior to the stage of overt nephropathy and requires further study in patients at greatest risk for diabetic nephropathy: those with uncontrolled diabetes and underlying genetic susceptibility.[37-40]

APPENDIX

The remaining members of the North American Microalbuminuria Study Group include: Louis Amorosa, MD, Robert Wood Johnson Medical School, New Brunswick, New Jersey; Jonathan R. Anolik, MD, Larchmont Medical Center, Mount Laurel, New Jersey; Stephen L. Aronoff, MD, Barbara Eubank, RN, Endocrine Associates of Dallas, PA, Dallas, Texas; Michael Bailie, MD, Lois Pelletier, RN, University of Connecticut Health Center, Farmington, Connecticut; Thomas Blevins, MD, Austin Diagnostic Research Center, Austin, Texas; M. Arthur Charles, MD, PhD, Monique Perry, RN, Western Diabetes Association, Garden Grove, California; Dana Clarke, MD, Donna Tomkey, RN, Diabetes Health Center, Salt Lake City, Utah; Daniel Einhorn, MD, Denise Humphries, RN, Cabrillo Hospital, San Diego, California; Alan J. Garber, MD, PhD, Houston, Texas; Burritt Haag, MD, Evelyn Williams, RN, Baystate Medical Center, Springfield, Massachusetts; Wendy Hoy, MB, BS, Margaret Watkins, RN, Lovelace Medical Foundation, Albuquerque, New Mexico; Dorothy Kahkonen, MD, Davida Kruger, RN, Henry Ford Hospital, Detroit, Michigan; Charles Kilo, MD, FACP, Kilo Diabetes and Vascular Research Foundation, St. Louis, Missouri; Lori Laffel, MD, Ellen Golden, RN, Joslin Diabetes Center, Boston, Massachusetts; Peter Lodewick, MD, Deborah Lovinsky, RN, Diabetic Care Center, Birmingham, Alabama; Jeffrey Mason, MD, Sharon Johnson, RN, Diabetes Care Center, Fresno, California; Janet B. McGill, MD, Lois Schmidt, RN, Washington University, St. Louis, Missouri; Philip Orlander, MD, Kay Sorenson, RN, The University of Texas Health Science Center of Houston, Houston, Texas; Philip Raskin, MD, Diane Hellenbrand, RN, University of Texas, Dallas, Texas; Sidney Rosenblatt, MD, FACP, The Newport Clinic Research Center, Irvine, California; Sherwyn Schwartz, MD, Joia Aufieri, RN, San Antonio, Texas; John P. Sheehan, MD, Margie Ulchaker, RN, Westlake, Ohio; Jay Samuel Skyler, MD, Robert Agramonte, RN, Jackson Medical Towers, Miami, Florida; Jerome Starr, MD, Cynthia Gagliastre, RN, Worcester Diabetes Metabolic Associates, Inc., Worcester, Massachusetts; Meng H. Tan, MD, Jean Tan, RN, Halifax Infirmary, Halifax, Nova Scotia, Canada; Terry Taylor, MD, Karen Dawn, RN, Georgetown University Hospital, Washington, DC; W. Kenneth Ward, MD, Marilyn Rigdon, RN, Portland Diabetes Center, PC, Portland, Oregon.

REFERENCES

1. Bojestig M, Arnqvist HJ, Hermansson G, et al. Declining incidence of nephropathy in insulin-dependent diabetes mellitus. *NEJM.* 1994;330:15–18.
2. Viberti GL, Jarrett RJ, Mahmud U, et al. Microalbuminuria as a predictor of clinical nephropathy in insulin dependent diabetes mellitus. *Lancet.* 1982;i:1430–1432.
3. Parving H-H, Oxenboll B, Svendsen P, et al. Early detection of patients at risk of developing diabetic nephropathy: a longitudinal study of urinary albumin excretion. *Acta Endocrinol.* 1982;100:550–555.
4. Mogensen CE, Christensen CK. Predicting diabetic nephropathy in insulin-dependent patients. *NEJM.* 1984;311:89–93.
5. Mathiesen ER, Oxenboll B, Johansen K, et al. Incipient nephropathy in type 1 (insulin-dependent) diabetes. *Diabetologia.* 1984;26:406–410.
6. Mogensen CE. Microalbuminuria as a predictor of clinical diabetic nephropathy. *Kidney Int.* 1987;31:673–689.
7. Feldt-Rasmussen B, Mathiesen ER, Deckert T. Effect of two years of strict metabolic control on progression of incipient nephropathy in insulin-dependent diabetes. *Lancet.* 1986;2:1300–1304.
8. The Diabetes Control and Complications Trial Research Group. The effect of intensive treatment of diabetes on the development and progression of long-term complications in insulin-dependent diabetes mellitus. *NEJM.* 1993; 329:977–986.
9. Reichard P, Nilson B-Y, Rosenqvist U. The effect of long-term intensified insulin treatment on the development of microvascular complications of diabetes mellitus. *NEJM.* 1993;329:304–309.
10. Mogensen CE. Long-term antihypertensive treatment inhibiting progression of diabetic nephropathy. *BMJ.* 1982;285:685–688.
11. Parving H-H, Andersen AR, Smidt UM, Svendsen PA. Early aggressive antihypertensive treatment reduces rate of decline in kidney function in diabetic nephropathy. *Lancet.* 1983;1:1175–1179.
12. Kasiske BL, Kalil RSN, Ma JZ, et al. Effect of antihypertensive therapy on the kidney in patients with diabetes: a meta-regression analysis. *Ann Intern Med.* 1993;118:129–138.
13. Marre M, Chatellier G, Leblanc H, et al. Prevention of diabetic nephropathy with enalapril in normotensive insulin dependent diabetic patients with microalbuminuria. *BMJ.* 1988;297:1092–1095.
14. Mathiesen ER, Hommel E, Giese J, Parving H-H. Efficacy of captopril in postponing nephropathy in normotensive insulin dependent diabetic patients with microalbuminuria. *BMJ.* 1991;303:81–87.
15. Lewis EJ, Hunsicker LG, Bain RP, Rohde RD, for the Collaborative Study Group. The effect of angiotensin-converting enzyme inhibition on diabetic nephropathy. *NEJM.* 1993;329:1456–1462.
16. Viberti GC, Mogensen CE, Groop LC, Pauls JF, for the European Microalbuminuria Captopril Study Group. Effect of captopril on progression to clinical proteinuria in patients with insulin-dependent diabetes mellitus and microalbuminuria. *JAMA.* 1994;271:275–279.
17. Wang PH, Lau J, Chalmers TC. Meta-analysis of effects of intensive blood glucose control on late complications of type I diabetes. *Lancet.* 1993;341:1306–1309.
18. Kaplan EL, Meier P. Nonparametric estimation from incomplete observations. *J Am Stat Assoc.* 1958;53:457–481.
19. Lee ET. *Statistical Methods for Survival Data Analysis.* 2nd ed. New York: Wiley & Sons; 1992:107–109.
20. Mogensen CE, Hansen KW, Pedersen MM, Christensen CK. Renal factors influencing blood pressure threshold and choice of treatment for hypertension in IDDM. *Diabetes Care.* 1991;14 (suppl 4):13–26.
21. Bjorck S, Nyberg G, Mulec H, et al. Beneficial effects of angiotensin converting enzyme inhibition on renal function in patients with diabetic nephropathy. *BMJ.* 1986;293:471–474.
22. Bjorck S, Mulec H, Johnsen SA, et al. Renal protective effect of enalapril in diabetic nephropathy. *BMJ.* 1992;304:339–343.
23. Hostetter TH, Rennke HG, Brenner BM: The case of intrarenal hypertension in the initiation and progression of diabetic and other glomerulopathies. *Am J Med.* 1982;72:375–380.
24. Zatz R, Dunn BR, Meyer TW, et al. Prevention of diabetic glomerulopathy by pharmacological amelioration of glomerular capillary hypertension. *J Clin Invest.* 1986;77:1925–1930.
25. Anderson S, Rennke HG, Garcia DL, Brenner BM. Short and long term effects of antihypertensive therapy in the diabetic rat. *Kidney Int.* 1989;36:526–536.
26. Gansevoort RT, de Zeeuw D, de Jong PE. Dissociation between the course

of the hemodynamic and antiproteinuria effects of angiotensin I converting enzyme inhibition. *Kidney Int.* 1993;44:579–584.
27. Chavers BM, Bilous RW, Ellis EN, et al. Glomerular lesions and urinary albumin excretion in type I diabetes without overt proteinuria. *NEJM.* 1989; 320:966–970.
28. Jenkins DAS, Cowan P, Collier A, et al. Blood glucose control determines the renal hemodynamic response to angiotensin converting enzyme inhibition in type I diabetes. *Diabetic Med.* 1990;7:252–257.
29. Krolewski AS, Laffel LMB, Krolewski M, et al. Glycosylated hemoglobin and the risk of microalbuminuria in patients with insulin-dependent diabetes mellitus. *NEJM.* 1995;332:1251–1255.
30. Mogensen CE. Diabetic Renal Disease: the quest for normal tension–and beyond. *Diabetic Med.* 1995;12:756–769.
31. Siegel JE, Krolewski AS, Warram JH, Weinstein MC. Cost-effectiveness of screening and early treatment of nephropathy in patients with insulin-dependent diabetes mellitus. *J Am Soc Nephrol.* 1992;3(suppl):S111–S119.
32. Borch-Johnsen K, Wenzel H, Viberti GC, Mogensen CE. Is screening and intervention for microalbuminuria worthwhile in patients with insulin dependent diabetes? *BMJ.* 1993;306:1722–1723.
33. Viberti GC, on behalf of the Microalbuminuria Collaborative Study Group, UK. Intensive therapy fails to prevent overt nephropathy in insulin dependent diabetics with microalbuminuria. *Diabetologia.* 1994;37:A196. Abstract.
34. Laffel L, Warram J, Krolewski A. Glycemic control does not predict progression from low to high microalbuminuria (MA) in insulin-dependent diabetes mellitus (IDDM). *J Am Soc Nephrol.* 1994;5:379. Abstract.
35. Almdal T, Norgaard K, Feldt-Rasmussen B, Deckert T. The predictive value of microalbuminuria in IDDM. *Diabetes Care.* 1994;17:120–125.
36. Forsblom CM, Groop PH, Ekstrand A, Groop LC. Predictive value of microalbuminuria in patients with insulin-dependent diabetes of long duration. *BMJ.* 1992;305:1051–1053.
37. Doria A, Warram JH, Krolewski AS. Genetic predisposition to diabetic nephropathy: evidence for a role of the angiotensin I-converting enzyme gene. *Diabetes.* 1994;43:690–695.
38. Marre M, Bernadet P, Gallois Y, et al. Relationships between Ang I converting enzyme gene polymorphism, plasma levels, and diabetic retinal and renal complications. *Diabetes.* 1994;43:384–388.
39. Krolewski AS, Canessa M, Warram JH, et al. Predisposition to hypertension and susceptibility to renal disease in insulin-dependent diabetes mellitus. *NEJM.* 1988;318:140–145.
40. Mangili R, Bending JJ, Scott G, et al. Increased sodium-lithium countertransport activity in red cells of patients with insulin-dependent diabetes mellitus and nephropathy. *NEJM.* 1988;318:146–150.